ABHAY DESHMUKH

ECE STUDENT

7057490137 | deshmukhabhay508@gmail.com | linkedin.com/in/abhay-deshmukh | Nagpur, Maharashtra, India

PROFESSIONAL SUMMARY

Final-year ECE student with hands-on experience programming and commissioning Allen Bradley and Siemens PLC platforms, designing ladder logic control schemes, and integrating industrial communication protocols (Modbus, OPC UA, MQTT) into manufacturing and monitoring systems. Skilled in SCADA & HMI configuration, process control, panel wiring, and field instrumentation, with applied experience in embedded systems and IoT-based condition monitoring for predictive maintenance.

TECHNICAL SKILLS

PLC Programming: Allen Bradley (Studio 5000, RSLogix 5000, Micrologix 1100), Siemens (TIA Portal, S7-1200 & S7-1500), Mitsubishi (GX Works2 & GX Works3)

Ladder Logic, Function Block Diagrams, Structured Text

SCADA & HMI: InTouch, Wonderware SCADA, DOPSoft HMI, HMI Design & Configuration

Industrial Communication: Modbus (RTU & TCP), OPC UA, Profibus & Profinet, MQTT, Industrial Networking

Control Systems: Transfer Functions, Closed-Loop Control, Process Control, VFD Configuration

Electrical & Instrumentation: Sensor & Actuator Interfacing, Field Instrumentation, Panel Wiring

Embedded Systems: Microcontrollers (ESP32, Arduino), Embedded C, Sensor Integration

Standards & Practices: Industry 4.0 Integration, Preventive & Predictive Maintenance, Safety Systems

KEY PROJECTS

Intelligent PLC-Based Industry 4.0 Smart Manufacturing System

- Designed an industrial automation framework combining Allen Bradley PLC control logic with OPC UA and MQTT for real-time machine data exchange.
- Built an LLM-assisted predictive maintenance layer using a RAG pipeline and ANN-based fault classifier, applying safety separation principles for human-machine interaction.
- Documented the full system in an IEEE-format technical paper.

IoT-Enabled Predictive Maintenance System

- Built an ESP32-based monitoring node with 5 sensors (temperature, vibration via ADXL345, pressure via MPX5700AP, humidity via DHT22, noise via MAX4466).
- Integrated sensor data with AWS IoT Core over MQTT; authored a complete hardware setup and deployment guide.
- Benchmarked ANN and ensemble ML models for fault detection; ANN delivered the best classification accuracy.

Automatic Car Parking System

- Designed and programmed an Automatic Car Parking System using an Allen Bradley MicroLogix 1100 PLC, with ladder logic controlling slot detection, gate operation, and vehicle counting.
- Integrated IR and proximity sensors for real-time slot occupancy status and automated entry and exit gate sequencing.

Automatic Bottle Filling Machine

- Developed a PLC-based ladder logic program to automate a bottle filling line, sequencing bottle detection, fill-level control, and conveyor indexing.
- Applied sensor-based interlocking and timing logic to ensure accurate fill levels and synchronized conveyor motion.

EXPERIENCE

Embedded Systems Intern — Pantech Prolabs India *20th Feb - 20th May*

- Completed hands-on training in Embedded C programming on microcontroller platforms, covering peripheral interfacing and firmware development.

EDUCATION

B.Tech, Electronics & Communication Engineering *April 2027*

Tulsiramji Gaikwad-Patil College of Engineering and Technology (TGPCET), Nagpur | Aggregate CGPA: 8.7 (Final Year)

HSC (12th) - Jawahar Navodaya Vidyalaya, Yavatmal

83.60%

SSC (10th) - Jawahar Navodaya Vidyalaya, Yavatmal

89.60%